3 Problems

3.1 Problem 1: Ted Slaman and Dariusz Kalociński (computability theory)

We consider infinite countable first-order structures, their automorphism groups, and the descriptive aspects of their representations. A representation $\mathcal{A}$ of such a structure is an instance of its isomorphism type in which the domain is the set of natural numbers $\mathbb{N}$, and the constants, functions, and relations are elements of $\mathbb{N}$, multi-variable functions from $\mathbb{N}$ to $\mathbb{N}$, and multi-place relations on $\mathbb{N}$, respectively. An automorphism of $\mathcal{A}$, or an isomorphism from $\mathcal{A}$ to $\mathcal{B}$, is a bijection of $\mathbb{N}$ preserving the evaluation of all constants, functions, and relations. We let $AUT(\mathcal{A})$ denote the automorphism group of $\mathcal{A}$.

Definition.

- $\mathcal{A}$ is *computably AUT-countable* if every automorphism of $\mathcal{A}$ is computable relative to $\mathcal{A}$.
- $\mathcal{A}$ is *computably AUT-countable on a cone* if there is a $C \subseteq \mathbb{N}$ such that whenever $\mathcal{B}$ is isomorphic to $\mathcal{A}$, every automorphism of $\mathcal{B}$ is computable relative to $\mathcal{B} \oplus C$.

Definition. A function $f\colon \mathcal{A} \to \mathcal{A}$ is Σ_1^{in}*-definable* in $\mathcal{A}$ relative to parameters $\bar{a} = (a_1, \ldots, a_n)$ from $\mathcal{A}$ if there is a family of existential formulas $(\varphi_i)_{i\in\mathbb{N}}$ such that for all $x, y \in \mathcal{A}$, $f(x) = y$ if and only if there is an $i \in \mathbb{N}$ with $\mathcal{A} \models \varphi_i(\bar{a}, x, y)$.

Problem. Is there a countable computably-represented structure $\mathcal{A}$ such that $\mathcal{A}$ is computably AUT-countable on a cone, and such that for any finite set of parameters $\bar{a}$ in $\mathcal{A}$ there is an automorphism π of $\mathcal{A}$ which is not Σ_1^{in}-definable in $\mathcal{A}$ relative to the parameters $\bar{a} \cup \pi(\bar{a})$, where $\pi(\bar{a})$ is the image of $\bar{a}$ under π?

Pre-registered Statement (*Ted Slaman and Dariusz Kalociński*).

A basic observation is that when the automorphism group of $\mathcal{A}$ is countable, there is a finite set of parameters in $\mathcal{A}$ such that every automorphism of $\mathcal{A}$ is determined by its action on that set. Intuitively, our question asks whether there is an $\mathcal{A}$ for which every automorphism can be effectively defined relative to some set of parameters in $\mathcal{A}$, but there is no single set of parameters such that every automorphism π can be effectively defined relative to that set and its set of images under π.

As it turns out, there is such a structure.

There are many paradigms to produce structures with countable automorphism groups, and also standard diagonal methods to build structures while controlling which functions are definable from which parameters within those structures. The novelty of the solution that we present is that it does not apply any of those diagonal methods. Instead, the example presented is a remix of the additive group on the rational numbers, and the proof that it has the desired properties is geometrically motivated. Once the pattern of argument is formulated, the verification is elementary.

It took us several days, but not weeks, to find this example.[3]

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[3]One of the referees pointed out that an expert who was sufficiently versed in the work of Turetsky and the work of Alvir–Greenberg–Harrison-Trainor–Turetsky could solve the problem in a few hours.

3.2 Problem 2: Richard Schwartz (discrete geometry)

Let $\mathbb{Z}$ denote the integers and let $\mathbb{R}$ denote the reals. Let $\Sigma = \mathbb{R} \times [0,1]$ and let $\partial\Sigma = \mathbb{R} \times \{0,1\}$ denote the boundary of Σ. For $\beta > 0$ define $G_\beta \colon \Sigma \to \Sigma$ by

$$G_\beta(x,y) = (x+\beta,\, 1-y).$$

Let G_β^k denote the k-fold composition of either G_β or its inverse G_β^{-1}, according as $k > 0$ or $k < 0$; when $k = 0$, G_β^k is the identity.

A *clean triangle* is the convex hull of 3 non-collinear points in $\partial\Sigma$. A *clean triangulation* of Σ is a countable and locally finite collection of clean triangles, having pairwise disjoint interiors, whose union is Σ. The clean triangulation is G_β*-invariant* if the action of G_β preserves the triangulation.

A *squeeze map* of a clean triangle τ is an affine map from τ into $\mathbb{R}^3$ which decreases the length of the edge of τ that is contained in $\partial\Sigma$ and increases the lengths of the other two edges.

Call $\beta > 0$ *realized* if there is a continuous map $f \colon \Sigma \to \mathbb{R}^3$ with the following properties:

1. f restricts to a squeeze map on each triangle of a G_β-invariant clean triangulation of Σ;
2. $f(p) = f(q)$ if and only if $p = G_\beta^k(q)$ for some $k \in \mathbb{Z}$ (for all $p, q \in \Sigma$).

Problem. Prove that $\sqrt{3}$ is the infimum of the set of realized numbers.

Pre-registered Statement (*Richard Schwartz*).

Problem context. The problem presented is a mathematical variant of the 1977 Halpern–Weaver Conjecture about paper Moebius bands. The question goes back at least to a 1962 paper by Wunderlich. I solved this conjecture several years ago (R. E. Schwartz, *The optimal paper Moebius band*, Annals of Math., 2025). The main result in my paper is that if a flat Moebius band has a smooth isometric embedding into $\mathbb{R}^3$ then its aspect ratio is greater than $\sqrt{3}$. It had already been known that there exist paper Moebius bands of aspect ratio $\sqrt{3}+\epsilon$ for any $\epsilon > 0$, so $\sqrt{3}$ is the sharp result. My solution is kind of unusual: it was an old conjecture, but the solution uses only undergraduate (or maybe beginning graduate) level mathematics. The difficulty with the conjecture is that people did not know how to use the topological hypothesis in a geometry problem.

Novelty. The problem I am asking the AI is different from the one I solved in my paper in an important way. Here, I formulate the problem in terms of continuous piecewise affine maps which are not even local isometries. One virtue of this formulation is that the solution can be given entirely in terms of affine and Euclidean geometry; there is no reliance on results from the literature about zero-curvature smooth surfaces. It is all self-contained. The explicit connection to Moebius bands is that the quotient Σ/G_β is a flat Moebius band. The problem is phrased so that the AI cannot use any of the results in my paper in an off-the-shelf way: it would either have to come up with its own proof, or else recognize that my paper is relevant, understand its proofs, and translate everything into a different category of maps.

Proof strategy. There are two halves. The first is showing that $\beta > \sqrt{3}$. The central insight is to prove, using a variant of the Borsuk–Ulam Theorem, that there exists a pair of straight line segments $A, B \subset \Sigma$ such that:

- both A and B have their endpoints in $\partial\Sigma$;
- the images $f(A)$ and $f(B)$ are perpendicular, coplanar, and have disjoint interiors;

- the restriction of f to each of A and B is a distance-increasing affine map.

Once this is in place, the idea (after A and B are suitably ordered) is to compare the geometry of the trapezoid bounded by A and $G_\beta(A)$ with the constraints forced by $f(A)$ and $f(B)$; the comparison is a simple but kind of miraculous optimization problem in planar Euclidean geometry. The second half involves an explicit construction showing each $\beta = \sqrt{3} + \epsilon$ can be realized, by correctly perturbing away from the so-called triangular Moebius band.

Difficulty. I would expect that a good graduate student (or a stellar undergraduate) in geometry and/or low-dimensional topology could solve this fairly quickly, knowing my solution to the Halpern–Weaver Conjecture. Without knowing my solution, one might work on this problem for years: the original problem was open for almost 50 years and I spent about 4 years (off and on) trying to solve it. If the AI finds a correct solution that does not resemble my own, I will be wildly impressed.

3.3 Problem 3: Aleksa Milojević and Benny Sudakov (discrete probability)

This problem and its solution will appear as part of the forthcoming paper:

- *Aleksa Milojević and Benny Sudakov, On the Probability a Weighted Bernoulli Sum Exceeds Its Mean, manuscript.*

Let $w_1, \ldots, w_m$ be non-negative real numbers such that $\sum_{i=1}^m w_i = 1$, and let $v_1, \ldots, v_m$ be i.i.d. Bernoulli(p) random variables. For which values of the probability p is it true that

$$\Pr\left[\sum_{i=1}^m w_i v_i \geq p\right] \geq p?$$

Pre-registered Statement (*Aleksa Milojević and Benny Sudakov*).

The main novelty of our argument is observing the connection of the probabilistic inequality $\Pr\left[\sum_{i=1}^m w_i v_i \geq p\right] \geq p$ with the Manickam–Miklós–Singhi (MMS) conjecture. This conjecture states that for any positive integers n, k with $n \geq 4k$, any set of n real numbers with nonnegative sum has at least $\binom{n-1}{k-1}$ subsets of size k with nonnegative sum. In particular, we observe that a solution to the MMS conjecture would imply this probabilistic inequality, and we use the results of Pokrovskiy resolving a weak form of the MMS to establish our inequality for $p \leq 10^{-46}$.

The reduction to the MMS conjecture is based on a coupling argument, which generalizes a previous argument of Alon, Emek, Feldman and Tennenholtz (which dealt only with the case $p = 1/n$ for $n \in \mathbb{Z}_{>0}$). Although nontrivial, this coupling argument is not the most important element of our proof—observing the connection to the MMS conjecture is.

It is therefore hard to judge precisely how hard the problem is: beyond a nice idea, one also needs wide-area knowledge to spot the connection mentioned above. Still, it is safe to say that this observation qualifies more as a nice note than as a fundamental breakthrough. That being said, we suspect that proving the inequality for all $p \leq 1/3$ [4] may be comparably hard as solving the MMS conjecture fully. This would be a very hard task, as witnessed by the fact that this almost 40-year old conjecture is still only partially resolved, despite a series of serious attempts.

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[4] $p \leq 1/3$ is the range in which one might suspect that inequality holds.

3.4 Problem 4: Larry Guth (metric geometry)

Suppose (M, g) and (N, h) are piecewise smooth Riemannian manifolds. If $f\colon M \to N$ is piecewise smooth, we define the k-dilation of f as the infimal Λ so that for every k-dimensional submanifold $\Sigma \subset M$,

$$\mathrm{Vol}_k(f(\Sigma)) \leq \Lambda \ \mathrm{Vol}_k(\Sigma).$$

We write the k-dilation of f as $\mathrm{Dil}_k(f)$. Equivalently, $\mathrm{Dil}_k(f) = \sup_{x \in M} \|\Lambda^k df\|$.

Proposition. Prove that there is a constant $k > 0$ so that the following holds. Suppose that $R \subset \mathbb{R}^4$ is a 4-dimensional rectangle with side lengths $R_1 \leq R_2 \leq R_3 \leq R_4$ and S is a 4-dimensional rectangle with side lengths $S_1 \leq S_2 \leq S_3 \leq S_4$. Suppose that $f\colon (R, \partial R) \to (S, \partial S)$ is a piecewise smooth map with degree 1 and with $\mathrm{Dil}_2(f) \leq 1$. Suppose that $R_1 \leq kS_1$. Then either

$$R_3 R_4 > kS_3 S_4, \qquad \text{or} \qquad R_1 R_2 R_3 R_4 > kS_1 S_2^{1/2} S_3^{3/2} S_4.$$

Pre-registered Statement (*Larry Guth*).

The problem asks to study the 2-dilation of degree-1 maps between 4-dimensional rectangles. More precisely, if R and S are the two rectangles, we are asking about degree-1 maps $f : (R, \partial R) \to (S, \partial S)$. There is a simpler problem that was solved in all dimensions, which asks for a subset $U \subset R$ and a degree-1 map from $(U, \partial U)$ to $(S, \partial S)$. The main novelty is a new way to distinguish these two similar problems.

To control f, we consider rectangles $\Sigma \subset S$ and their inverse images $f^{-1}(\Sigma)$. We note that f maps the complement of $f^{-1}(\Sigma)$ to the complement of Σ. These complements have additional topology. We can map the complement of Σ to a wedge of spheres, and the resulting composition has non-trivial rational homotopy invariants. These can be defined using differential forms and related to the 2-dilation of f. The connection between rational homotopy invariants and differential forms is a technique in the literature, but the topology of this situation is a little different from previous ones. In this setting, the argument leads to a lower bound on the 3-volume of any chain with boundary $f^{-1}(\Sigma)$. Using the isoperimetric profile of R (proven in previous work), this leads to a lower bound on the area of $f^{-1}(\Sigma)$, which leads to the given estimates.

The proposition came from an unpublished paper that I worked on as a postdoc. The paper had three somewhat novel steps, of which this is the first. This particular proposition probably took about two weeks of work, during which I spent about a third of my work time on it.

3.5 Problem 5: Oleg Butkovsky, Jonathan C. Mattingly, and Lorenzo Zambotti (stochastic PDE)

This problem and its solution will appear as part of the forthcoming paper:

- *Oleg Butkovsky Jonathan C. Mattingly Lorenzo Zambotti, manuscript in preparation, 2026.*

Let $u(x, t)$ be a real-valued solution to the following stochastic partial differential equation

(SPDE):

$$\begin{aligned}\frac{\partial u}{\partial t}(x,t) &= \frac{1}{2}\frac{\partial^2 u}{\partial x^2}(x,t) + \delta_0(u(x,t)) + \dot{W}(x,t) \quad \text{for } t \geq 0,\\ u(x,0) &= u_0(x) \in C_0([0,1]),\\ u(0,t) &= u(1,t) = 0 \quad \text{for all } t \geq 0,\end{aligned}$$

where $C_0([0,1]) := \{f \in C([0,1],\mathbb{R}) : f(0) = f(1) = 0\}$, $x \in [0,1]$, $t \geq 0$, δ_0 is the Dirac delta function at 0, $\dot{W}(x,t)$ is space-time white noise, and $u_0 \in C_0([0,1])$ is the initial condition. We understand the probabilistically weak PDE mild solution of the equation in the sense described in Athreya, Butkovsky, Lê, and Mytnik, "Well-posedness of stochastic heat equation with distributional drift and skew stochastic heat equation" (2024) [ABLM24]. We assume that the existence, uniqueness, and approximation results of [ABLM24] extend to the SPDE on $[0,1]$ with Dirichlet boundary conditions, i.e. with the Dirichlet heat kernel and initial data $u_0 \in C_0([0,1])$; in particular, that there exists a unique (in law) adapted mild solution on $C([0,T], C_0([0,1]))$.

We further assume that if u^n is a solution to this SPDE with the drift $G_{1/n}$ in place of δ_0, where $G_{1/n}(x) := \sqrt{\frac{n}{2\pi}}\exp(-\frac{nx^2}{2})$ is the 1D heat kernel on $\mathbb{R}$ with variance $1/n$, then for any $t > 0$ we have $u^n(t) \to u(t)$ weakly as $n \to \infty$.

Let P_t be the Markov semigroup associated with the solution $u(t)$, i.e. for any bounded measurable $f\colon C_0([0,1]) \to \mathbb{R}$,

$$P_t f(u_0) = \mathbb{E}\Big[f(u(\cdot,t)) \mid u(\cdot,0) = u_0(\cdot)\Big], \quad u_0 \in C_0([0,1]).$$

Problem. Prove or disprove: P_t has at most one invariant probability measure μ on $C_0([0,1])$.

Pre-registered Statement (*Jonathan Mattingly*).

Novelty. There are a number of classical methods—Doob's Theorem using the Strong Feller property, Dirichlet form methods, or coupling using Girsanov's Theorem—that one might hope to apply to prove unique ergodicity. The difficulty is that the drift term, which contains a delta function at zero, is not regular enough to apply those methods; the regularization due to the Laplacian is not sufficient. We apply asymptotic Strong Feller / asymptotic coupling methods to prove the theorem. Those applications are not straightforward either, again due to the irregularity of the drift.

Key references. "Skew heat equation" (2011) by Bounebache–Zambotti shows the existence of the invariant measure but comments that proving the Strong Feller property seems out of reach. "Well-posedness of stochastic heat equation with distributional drift and skew stochastic heat equation" (2023) by Athreya, Butkovsky, Lê, and Mytnik extends the results but still does not give the tools to prove uniqueness. We needed to include some estimates from a stochastic sewing lemma, which not standard in this domain.

Difficulty. We first discussed this problem in October 2025 as a group, building on work over the preceding years by Butkovsky and Zambotti, and had the proof done by January 2026. In total it took 4-5 weeks. However, this also required the right mix of people: some had the ergodicity proof idea, while others knew how to prove the needed regularity estimates.

3.6 Problem 6: Joshua Greene and Duncan McCoy (lattice theory)

This problem and its solution will appear as part of the forthcoming paper:

- *Joshua Evan Greene and Duncan McCoy, On the Neumann-Zagier conjecture, 2026, in preparation.*

Let T denote a finite, weighted tree with vertex set V, edge set E, and weight function $w\colon V \to \mathbb{Z}$. Let $L(T)$ denote the free abelian group generated by V, equipped with the symmetric bilinear form $\cdot$ whose value on a pair of generators $u, v \in V$ is given by

$$u \cdot v = \begin{cases} w(u), & u = v; \\ -1, & (u,v) \in E; \\ 0, & u \neq v,\ (u,v) \notin E. \end{cases}$$

Suppose that the form is positive definite, and suppose that there exists a single vertex v for which the vertex weight $w(v)$ is less than the vertex degree $d(v)$. Prove that T contains a vertex u such that u is an irreducible element of $L(T)$, i.e. there do not exist nonzero lattice elements $a, b \in L(T)$ such that $u = a + b$ and $a \cdot b \geq 0$.

Pre-registered Statement (*Joshua Evan Greene and Duncan McCoy*).

(i) We believe the argument is rather novel. There is not much literature on the Neumann–Zagier conjecture. The closest related paper is one of Issa and McCoy; however, our argument is very different from the argument in that paper. There are several subtle steps in our solution which depart from a straightforward attempt.

(ii) The general strategy of the proof is to argue that either the bad vertex is irreducible, or else there is a leaf of weight one (which is therefore irreducible). However, we make more delicate arguments and estimates, drawing on properties of the dual lattice: some known, some novel. Our submitted solution is laid out somewhat in reverse. Lemma 2.6 shows that if a vertex in a definite tree lattice is reducible, then some associated dual element is either reducible or else is contained in the lattice itself. We apply this to the bad vertex, the one whose weight is lower than its degree. Corollary 2.3 rules out the possibility that any associated dual element is irreducible; this is a quite novel argument, which sensitively uses the tree structure, and which backs onto an estimate for the lengths of elements in dual cut lattices, going back to Kirchhoff. Lemma 2.2 rules out the possibility that any associated dual element is contained in the lattice itself, unless there is a leaf of weight one.

(iii) We made an attempt on the Neumann–Zagier conjecture over the course of Summer 2020, getting the special case in which the tree lattice is isometric to the standard Euclidean lattice. We made another effort over a few weeks during Fall 2023, in which we realized we could prove the stronger result. All told, it was a serious effort by both of us over the course of several (4–5) months.

3.7 Problem 7: Sucharit Sarkar (combinatorial topology)

This problem and its solution will appear as part of the forthcoming paper:

- *Robert Lipshitz, Lenhard Ng, and Sucharit Sarkar, A stable homotopy refinement of Legendrian contact homology, in preparation.*

Let $G = \langle a_1, \dots, a_r \rangle$ be a free group. A word on the letters $L = \{a_1, a_1^{-1}, \dots, a_r, a_r^{-1}\}$ is called *reducible* if it equals $1 \in G$. A word is called *quasi-reduced* if it does not contain three consecutive letters $\ell\ell^{-1}\ell$ for any $\ell \in L$.

Fix a reducible quasi-reduced word $w = w_1 \cdots w_{2n}$ (with $w_i \in L$). A *matching* M consists of n intervals, each properly embedded in $\mathbb{R} \times \mathbb{R}_+$, satisfying:

- any two intervals intersect transversally, and there are no triple points;
- $\partial M = \{1, \dots, 2n\} \times 0 \subset \mathbb{R} \times 0 \subset \mathbb{R} \times \mathbb{R}_+$;
- for each component R of $\mathbb{R} \times \mathbb{R}_+ \setminus M$, and for each component A of $M \cap \partial\overline{R}$, ∂A consists of some two points $(i, 0)$ and $(j, 0)$ with $w_i = w_j^{-1}$.

A *k-crossing matching* is a matching where the total number of self-intersections is k.

Let F_w be the following CW complex whose k-cells correspond to isotopy classes of k-crossing matchings. Given a k-crossing matching M with crossing set $X(M)$, the associated k-cell is $C(M) := \prod_{c \in X(M)} I_c$, where I_c is an interval whose endpoints correspond to the two ways of locally resolving M near c. The attaching maps are as follows: on a facet of the form

$$a \times \prod_{d \in X(M) \setminus \{c\}} I_d, \qquad c \in X(M),\ a \in \partial I_c,$$

let M_a be the $(k-1)$-crossing matching obtained by resolving M at c according to a. Then the $(k-1)$-cell associated to M_a is $C(M_a) = \prod_{d \in X(M_a)} I_d = \prod_{d \in X(M) \setminus \{c\}} I_d$, and the above facet maps to this $(k-1)$-cell canonically.

Problem. Is F_w contractible?

Pre-registered Statement (*Sucharit Sarkar*).

The novelty lies in the definition of the CW complex F_w. While similar definitions have appeared elsewhere (like matching complexes), this specific definition of F_w appears to be completely new. (In the actual paper, we study an even more general complex, where the restrictions "quasi-reduced" and "reducible" on w are dropped.) The challenge is in "comprehending" this new concept. Once one understands what the definition actually represents, the properties of F_w are fairly easy to deduce. For instance, F_w is always 1-connected—it took the authors around 1 hour to prove that; and then they tried desperately to prove that F_w is 2-connected as well. After around 2–3 hours of unsuccessfully trying to adapt the previous proof, they tried to find a counterexample, which they then immediately found (much to their disappointment).

3.8 Problem 8: Sam Payne and Jidong (Jayden) Wang (matroids and tropical geometry)

This problem and its solution will appear as part of the forthcoming paper:

- *Jidong Wang, in preparation, 2026.*

Let M be a matroid, and let $\mathrm{Trop}(M)$ be the corresponding tropical linear space, i.e. the closure of the Bergman fan of M in tropical projective space. The *relative Dressian* of M is

$$\mathrm{Dr}(M) := \{\text{tropical linear subspaces of } \mathrm{Trop}(M)\},$$

which is partially ordered by inclusion of tropical linear spaces. Suppose M admits an order-reversing involution $F \mapsto F^\perp$ on the lattice of flats $\mathcal{L}(M)$. Note that there is a natural inclusion $\mathcal{L}(M)^{\mathrm{op}} \subset \mathrm{Dr}(M)$ given by $F \mapsto \mathrm{Trop}(M/F \oplus U_{0,F})$.
Problem. Is there an order-reversing involution Φ on $\mathrm{Dr}(M)$ that extends the involution $F \mapsto F^\perp$?

Pre-registered Statement (*Sam Payne and Jidong (Jayden) Wang*).

This problem is at the level of an interesting technical proposition. We expect that an advanced graduate student working on matroid theory and tropical linear spaces would be able to find a solution. The proof requires three steps: (1) understanding that the answer should be affirmative; (2) constructing the map, either directly as a tropical linear map on the ambient tropical projective spaces containing the corank-k Dressians in their respective Plücker embeddings, or by passing to a cryptomorphic characterization of valuated quotients as functions on lattices (as in our solution); (3) verifying that this map is well-defined, i.e. that it takes points in the corank-k Dressian to points in the rank-k Dressian.

We estimate that a sufficiently advanced AI system, by current standards, should solve (1) and (2) with probability 60–70%, but giving a clear and correct verification (step (3)) may be more challenging, with a probability of success of 40–50%.

3.9 Problem 9: Sylvie Corteel and John Lentfer (algebraic combinatorics)

This problem and its solution will appear as part of the forthcoming paper:

- *Sylvie Corteel and John Lentfer, Universal Hilbert series coefficients of the superspace coinvariant ring, 2026, in preparation.*

Let m and n be positive integers and let $R_n^{(m)}$ be the coinvariant algebra with m sets of n variables. The Hilbert series of an m-multigraded algebra A is

$$\mathrm{Hilb}(A; q_1, \ldots, q_m) = \sum_{i_1,\ldots,i_m \geq 0} \dim\left(A_{(i_1,\ldots,i_m)}\right) q_1^{i_1} \cdots q_m^{i_m},$$

where $A_{(i_1,\ldots,i_m)}$ denotes a multigraded component, with grading variables $q_1, \ldots, q_m$. Let $s_\lambda(q_1, \ldots, q_m)$ denote a Schur polynomial. For $R_n^{(m)}$, it is known that its Hilbert series may be expressed by

$$\mathrm{Hilb}(R_n^{(m)}; q_1, \ldots, q_m) = \sum_\lambda c_\lambda(n)\, s_\lambda(q_1, \ldots, q_m),$$

for some nonnegative universal series coefficients $c_\lambda(n)$ which do not depend on m.

Problem. For any n, give a combinatorial interpretation of the coefficients $c_\lambda(n)$ when λ is a hook shape partition.

Pre-registered Statement (*Sylvie Corteel and John Lentfer*).

Novelty. The definition of $R_n^{(m)}$ implies that if you know its Hilbert series and compute the Schur expansion, you get the coefficient $c_\lambda(n)$ for all partitions λ with at most m parts. For now this is known for $m = 1$, and the Hilbert series is known for $m = 2$ but not its Schur expansion. The only known hook shape is $\lambda = (a, 1^b)$ with $b = 0$; Opus, Gemini and ChatGPT know this result.

Strategy of the proof. The "trick" here is due to a result of John Lentfer during his PhD at Berkeley: if you compute the Hilbert series of the coinvariant algebras in m sets of commuting variables and j sets of anticommuting variables, and know their super Schur expansion, then you get the coefficient $c_\lambda(n)$ for any n and any λ with $\lambda_{m+1} \leq j$.

Difficulty. Once you know the trick, the hook shape corresponds to $m = j = 1$. The proof then uses a sign-reversing involution of Sagan and Swanson together with super Schur polynomials. Once you mix all these ingredients, the proof is not too long.

3.10 Problem 10: Srivatsav Kunnawalkam Elayavalli (von Neumann algebras)

The solution of this problem (which appears in the auxiliary solutions document) was written by

- *Srivatsav Kunnawalkam Elayavalli and Zhiyuan Yang*

Problem. Let Γ be a finite simplicial graph with $|V(\Gamma)| \geq 3$, and let (M_v, φ_v) be a separable tracial von Neumann algebra for each $v \in V$. If Γ is irreducible (in the sense that there do not exist subgraphs Γ_1, Γ_2 such that $V(\Gamma_1) \cup V(\Gamma_2) = V(\Gamma)$ and $(v, u) \in E(\Gamma)$ for all $v \in V(\Gamma_1), u \in V(\Gamma_2)$), and each M_v has a trace-zero unitary, then the graph product von Neumann algebra M is properly proximal.

Pre-registered Statement (*Srivatsav Kunnawalkam Elayavalli and Zhiyuan Yang*).

Novelty of the argument. The argument is a von Neumann algebraic analogue of the argument in the case of graph products of groups, as executed in the proof of Theorem 1.8 of the work of Ding and Kunnawalkam Elayavalli (2024). However, the implementation is difficult, because it has to be done in the intricate language of small-at-infinity compactifications of von Neumann algebras developed by Ding, Kunnawalkam Elayavalli and Peterson. Various subtleties arise in this setting that require nontrivial arguments: a fact well known to experts is that not every group-theoretic argument generalizes immediately to von Neumann algebras. For

instance, the results of Popa and Johnson–Parrot rule out the existence of derivations of a von Neumann algebra into the compact operators, despite the fact that plenty of these exist in the group-theoretic analogue.

General strategy of proof. The proof follows a "patching" argument, available thanks to the decompositional structure of graph products. Knowing that the graph is irreducible allows us to split the graph product into various nontrivial amalgamated free products, where each amalgam is the star of a fixed vertex. This yields "relative proper proximality" of the entire algebra with respect to the collection of subalgebras generated by the star subgraphs—an extension of the paradoxical decomposition trick that proves proper proximality for free products. The next part is more subtle, and involves upgrading this relative proper proximality, by proving that the support projections of the associated boundary pieces commute with each other and that their infimum is the compact operators. This is proved by combinatorially exploiting the Fock space representation of graph products (Caspers–Fima).

Estimate of difficulty. Being able to carry out a proof of this result would be a good marker for a strong third- or fourth-year graduate student working in this subfield of von Neumann algebras. I expect they would take a couple of weeks (two or three) to nail down and write up the argument in full detail.

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4 AI Systems Tested

The four systems below are listed in the order that their participation in the benchmark was confirmed.

4.1 System A: IMProofBench ProofCouncil

System A, called ProofCouncil, was developed by a team consisting of Johannes Schmitt (ETH Zürich), Tim Gehrunger (ETH Zürich), Jasper Dekoninck (ETH Zürich), Gergely Bérczi (Aarhus University), Uri Kreitner (ETH Zürich), and Liam Price (Independent Researcher). The system primarily used `gpt-5.5 pro` as its base model, though it also called `gpt-5.5`, `gemini-3.1-pro-preview`, and `claude-opus-4-7`. The source code for System A is available at `batch-2-submissions/improofbench`.

4.2 System B: UCLA Moonshot Harness

System B, called the UCLA Moonshot Harness, was developed by Junyi Zhang[5], Xinjie He[5], Hyunsik Chae, Ethan Ji, Eric Jiang, Rushil Raghavan, Yiwen Kou, Alex Taylor, Kai-Wei Chang[6], Raghu Meka[6], Violet Peng[6], Amit Sahai[6], Terence Tao[6], and Wei Wang[6], all of the University of California, Los Angeles. The system used `gpt-5.5 pro` as its base model. The source code for System B is available at `batch-2-submissions/ucla`.

[5]Co-first authors with equal contribution. The remaining students are ordered by contribution to the harness.

[6]Principal investigators, listed at the end in alphabetical order by last name.